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Joint lab on Safety and Security of AI

# Computer Vision Technologies and Biometrics
# Iris


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# Outline of the talk

- **Iris**
- **Characteristics**
- **Pre-processing**
- **Feature extraction**
- **Matching**

# History of iris as a recognition mean



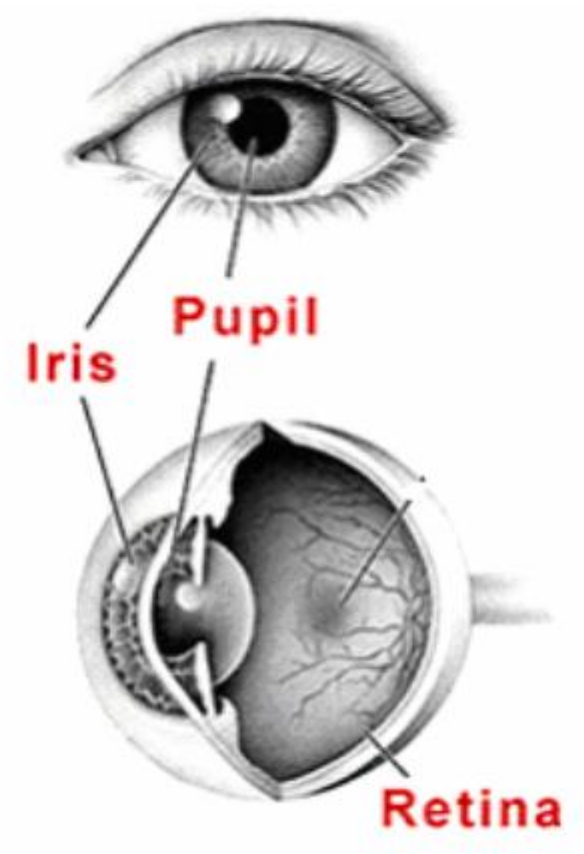
- **Iris for personal recognition between 1935 and 1939**
  - Carleton Simon, Isadora Goldstein, Frank Burch, Leonard Flom, Aran Safir

- **Early use of iris recognition was to distinguish inmates in the Parisian penal system**

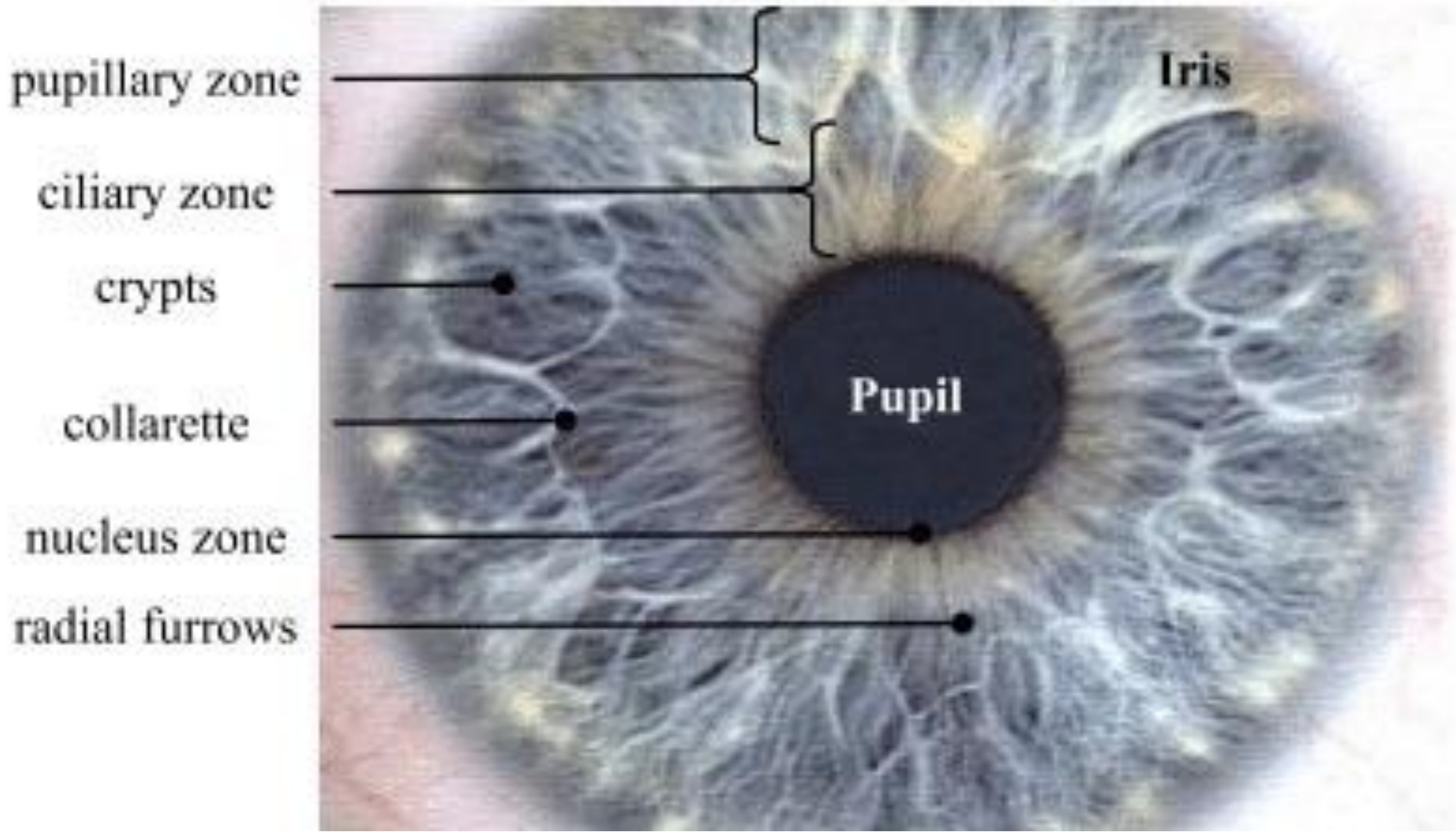
- **More recently, the concept of *automated* iris recognition was proposed**
  - John Daugman, 1991

- **Early work toward realizing a system for automated iris recognition was carried out at Los Alamos National Laboratories in the USA**
  - An IRIS group at Los Alamos is still working but has nothing to do with these early studies

- **Several research groups developed and documented prototype iris recognition systems working with highly cooperative subjects at *close* distances**

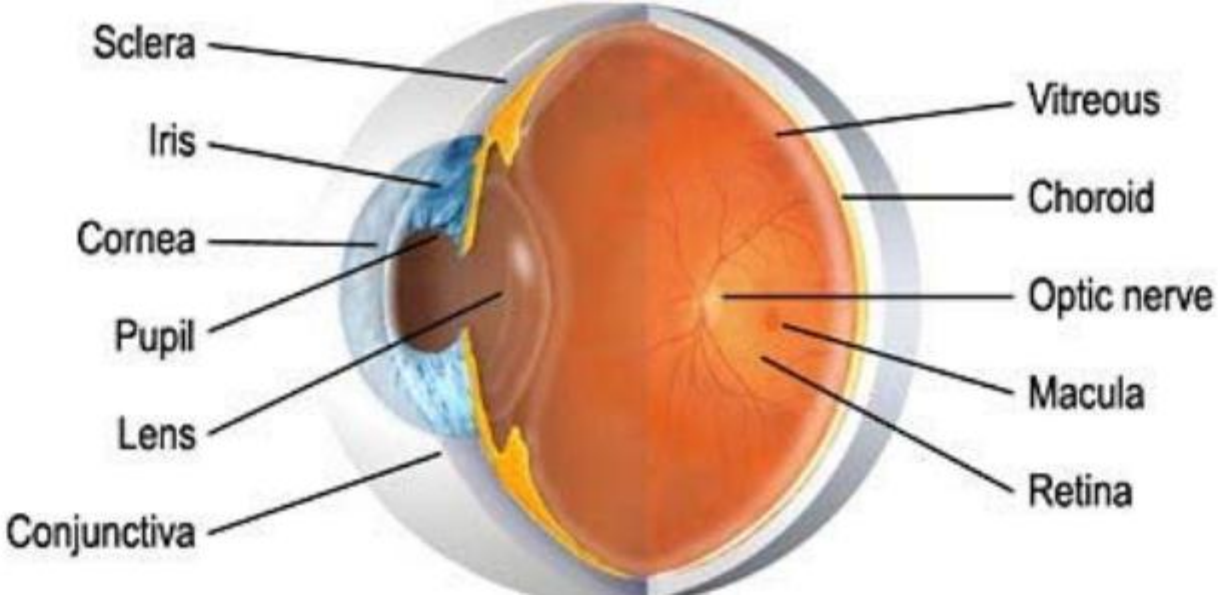


# The iris as biometric trait



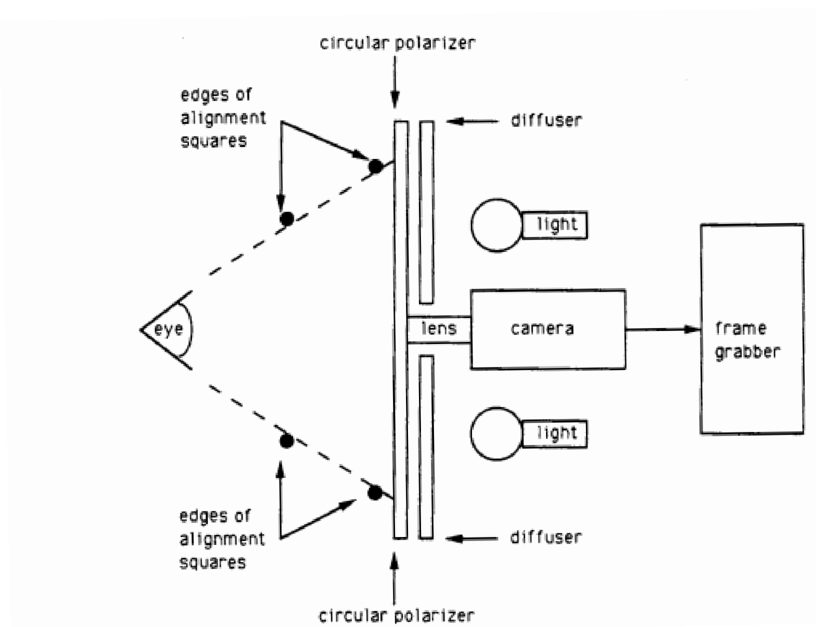
# Iris vs. retinal recognition

- **Iris recognition utilizes the iris muscle**
- **Retinal recognition uses the unique pattern of blood vessels on an individual's retina at the back of the eye**
- **Both techniques use NIR (near infrared) light with a power such that to prevent eye damages**



# Early work

- **R.P. Wildes, Iris recognition: an emerging technology, Proceedings of IEEE, 1997**




## Iris Recognition: An Emerging Biometric Technology


RICHARD P. WILDES, MEMBER, IEEE



*This paper examines automated iris recognition as a biometrically based technology for personal identification and verification. The motivation for this endeavor stems from the observation that the human iris provides a particularly interesting structure on which to base a technology for noninvasive biometric assessment. In particular, the biomedical literature suggests that irises are as distinct as fingerprints or patterns of retinal blood vessels. Further, since the iris is an overt body, its appearance is amenable to remote examination with the aid of a machine vision system. The body of this paper details issues in the design and operation of such systems. For the sake of illustration, extant systems are described in some amount of detail.*



***Keywords*—*Biometrics, iris recognition, machine vision, object recognition, pattern recognition.***


### I. Introduction

#### A. Motivation

Technologies that exploit biometrics have the potential for application to the identification and verification of individuals for controlling access to secured areas or materials.[1] A wide variety of biometrics have been marshaled in support of this challenge. Resulting systems include those based on automated recognition of retinal vasculature, fingerprints, hand shape, handwritten signature, and voice [24], [40]. Provided a highly cooperative operator, these approaches have the potential to provide acceptable performance. Unfortunately, from the human factors point of view, these methods are highly invasive: Typically, the operator is required to make physical contact with a sensing device or otherwise take some special action (e.g., recite a specific phonemic sequence). Similarly, there is little potential for covert evaluation. One possible alternative to these methods that has the potential to be less invasive is automated face recognition. However, while automated face recognition is a topic of active research, the inherent difficulty of the problem might prevent widely applicable technologies from appearing in the near term [9], [45]. Automated iris recognition is yet another alternative for noninvasive verification and identification of people. Interestingly, the spatial patterns that are apparent in the human iris are highly distinctive to an individual [1], [34] (see, e.g., Fig. 1). Like the face, the iris is an overt body that is available for remote (i.e., noninvasive) assessment. Unlike the human face, however, the variability in appearance of any one iris might be well enough constrained to make possible an automated recognition system based on currently available machine vision technologies.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The author is with The Sarnoff Corporation, Princeton, NJ 08543-5300.



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[1] Throughout this discussion, the term "verification" will refer to recognition relative to a specified data base entry. The term "identification" will refer to recognition relative to a larger set of alternative entries.

#### B. Background

The word *iris* dates from classical times (ιρις, a rainbow). As applied to the colored portion of the exterior eye, iris seems to date to the sixteenth century and was taken to denote this structure's variegated appearance [50]. More technically, the iris is part of the uveal, or middle, coat of the eye. It is a thin diaphragm stretching across the anterior portion of the eye and supported by the lens (see Fig. 2). This support gives it the shape of a truncated cone in three dimensions. At its base, the iris is attached to the eye's cilliary body. At the opposite end, it opens into the pupil, typically slightly to the nasal side and below center. The cornea lies in front of the iris and provides a transparent protective covering.

To appreciate the richness of the iris as a pattern for recognition, it is useful to consider its structure in a bit more detail. The iris is composed of several layers. Its posterior surface consists of heavily pigmented epithelial cells that make it light tight (i.e., impenetrable by light). Anterior to this layer are two cooperative muscles for controlling the pupil. Next is the stromal layer, consisting of collagenous connective tissue in arch-like processes. Coursing through this layer are radially arranged corkscrew-like blood vessels. The most anterior layer is the anterior border layer, differing from the stroma in being more densely packed, especially with individual pigment cells called chromataphores. The visual appearance of the iris is a direct result of its multilayered structure. The anterior surface of the iris is seen to be divided into a

# The father of Iris Recognition: John Daugman (born in 1954)

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- University of Cambridge, Computer Laboratory
- Patent 1991
- He is still continuing his research

This paper is his MILESTONE



## High Confidence Visual Recognition of Persons by a Test of Statistical Independence

John G. Daugman

*Abstract*—**A method for rapid visual recognition of personal identity is described, based on the failure of a statistical test of independence. The most unique phenotypic feature visible in a person's face is the detailed texture of each eye's iris: An estimate of its statistical complexity in a sample of the human population reveals variation corresponding to several hundred independent degrees-of-freedom. Morphogenetic randomness in the texture expressed phenotypically in the iris trabecular meshwork ensures that a test of statistical independence on two coded patterns originating from different eyes is passed almost certainly, whereas the same test is failed almost certainly when the compared codes originate from the same eye. The visible texture of a person's iris in a real-time video image is encoded into a compact sequence of multi-scale quadrature 2-D Gabor wavelet coefficients, whose most-significant bits comprise a 256-byte "iris code." Statistical decision theory generates identification decisions from Exclusive-OR comparisons of complete iris codes at the rate of 4 000 per second, including calculation of decision confidence levels. The distributions observed empirically in such comparisons imply a theoretical "cross-over" error rate of one in 131 000 when a decision criterion is adopted that would equalize the false accept and false reject error rates. In the typical recognition case, given the mean observed degree of iris code agreement, the decision confidence levels correspond formally to a conditional false accept probability of one in about $10^{31}$.**

*Index Terms*—**Image analysis, statistical pattern recognition, biometric identification, statistical decision theory, 2-D Gabor filters, wavelets, texture analysis, morphogenesis.**

### I. Introduction

EFFORTS to devise reliable mechanical means for biometric personal identification have a long and colorful history. In the Victorian era for example, inspired by the birth of criminology and a desire to identify prisoners and malefactors, Sir Francis Galton F.R.S. [13] proposed various biometric indices for facial profiles which he represented numerically. Seeking to improve on the system of French physician Alphonse Bertillon for classifying convicts into one of 81 categories, Galton devised a series of spring-loaded "mechanical selectors" for facial measurements and established an Anthropometric Laboratory at South Kensington [13]. Other biometric identifiers that have been adopted historically, ranging from cranial dimensions to digit length, as well as some of the numerous geometric facial measurements currently being tried, are described in [17], [25].

Today there is renewed interest in reliable, rapid, and unintrusive means for automatically recognizing the identity of persons. Security breaches in access to restricted areas at airports are known to have contributed to terrorism; and credit card fraud now costs six billion dollars annually [3]. Other applications for high confidence personal identification include passport control, bank automatic teller machines, protected access to premises or assets, law enforcement, government intelligence, entitlement verification, birth certificates, licenses, and any existing use of keys or cards. Some of the identifying biometric features now under investigation for potential use include hand geometry, blood vessel patterns in the retina or hand, fingerprints, voice-prints, and handwritten signature dynamics. The critical attributes for any such measure are: the number of degrees-of-freedom of variation in the chosen index across the human population, since this determines uniqueness; its immutability over time and its immunity to intervention; and the computational prospects for efficiently encoding and reliably recognizing the identifying pattern.

The possibility that the iris of the eye might be used as a kind of optical fingerprint for personal identification was suggested originally by ophthalmologists [1], [12], [24], who noted from clinical experience that every iris had a highly detailed and unique texture, which remained unchanged in clinical photographs spanning decades (contrary to the occult diagnostic claims of "iridology"). Among the visible features in an iris, some of which may be seen in the close-up image of Fig. 1, are the trabecular meshwork of connective tissue (pectinate ligament), collagenous stromal fibres, ciliary processes, contraction furrows, crypts, a serpentine vasculature, rings, corona, coloration, and freckles [1], [11], [12], [24]. The striated trabecular meshwork of chromatophore and fibroblast cells creates the predominant texture under visible light [24], but all of these sources of radial and angular variation taken together constitute a distinctive "fingerprint" that can be imaged at some distance from the person. Further properties of the iris that enhance its suitability for use in automatic identification include 1) its inherent isolation and protection from the external environment, being an internal organ of the eye, behind the cornea and the aqueous humor; 2) the impossibility of surgically modifying it without unacceptable risk to vision; and 3) its physiological response to light, which provides a natural test against artifice.

A property the iris shares with fingerprints is the random morphogenesis of its minutiae. Because there is no genetic

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IEEE Log Number 9212305.



Invited lecture: https://www.youtube.com/watch?v=KyDoFrojEYk

# Why the Iris?

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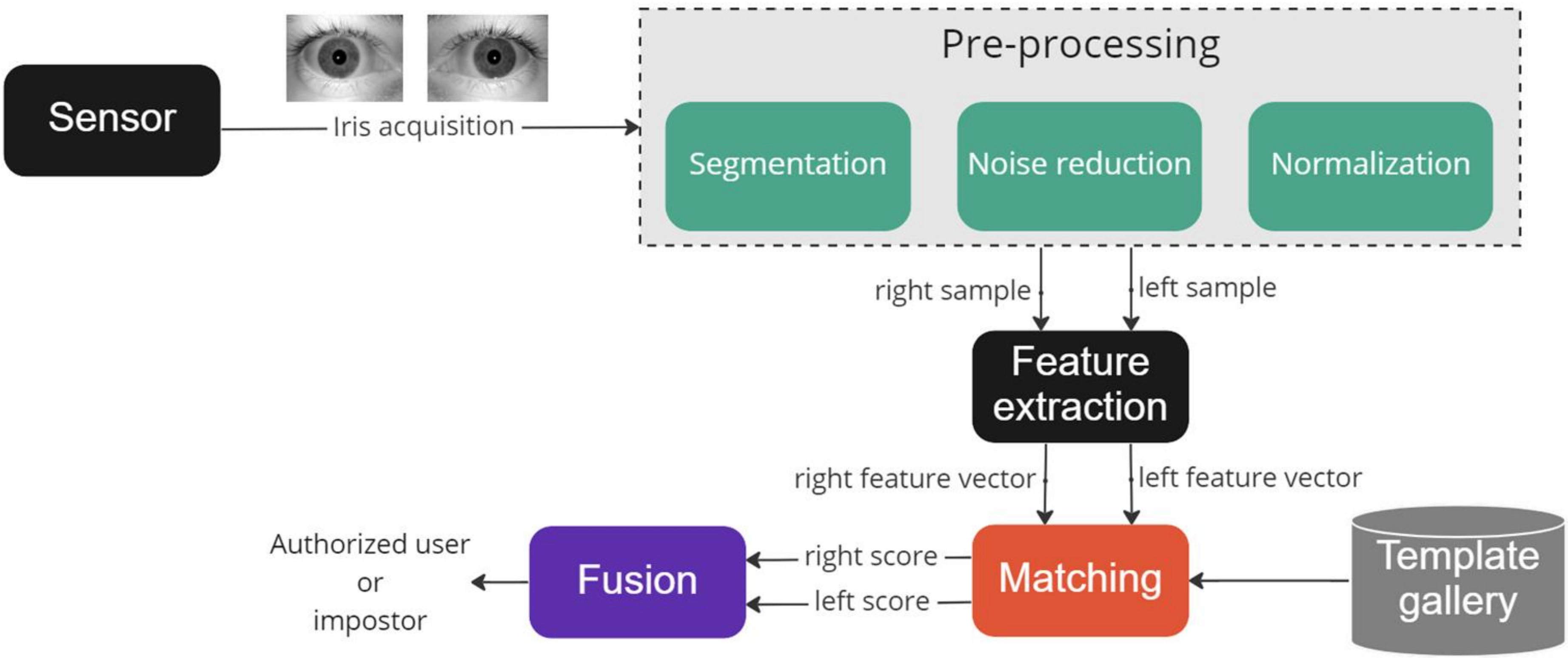
- **Non-invasive**
- **Contactless**
- **Uniqueness**
- **Invariance over time**
- **Stable characteristics**
- **Robust to spoofing**

# An Iris Recognition System



# Iris recognition steps

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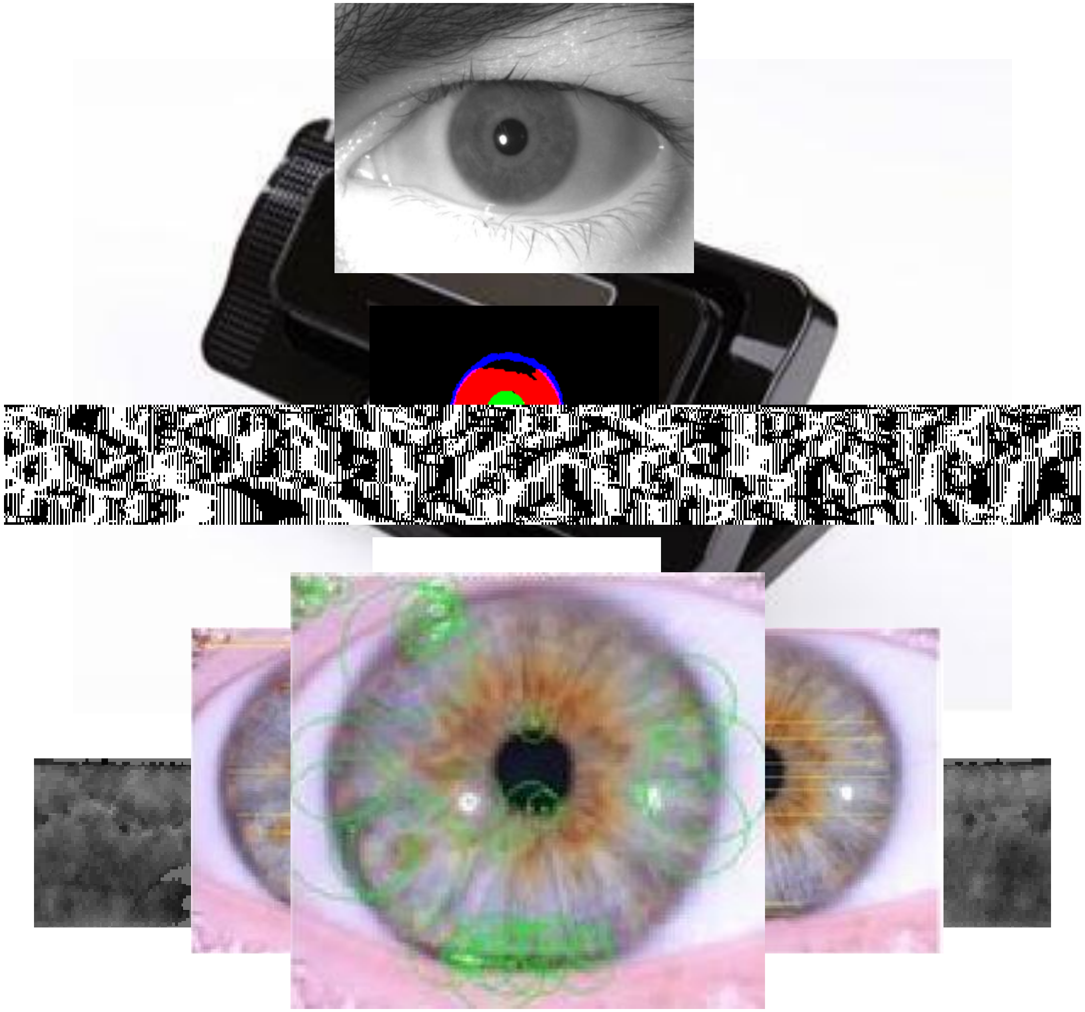
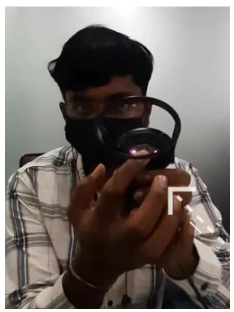
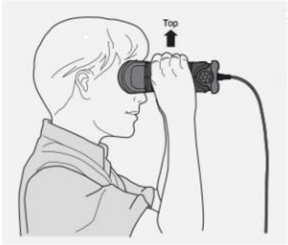
- Capture device
- Pre-processing
- Feature extraction

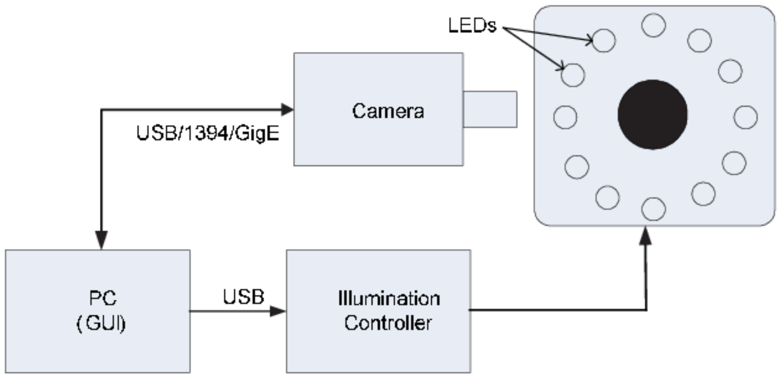


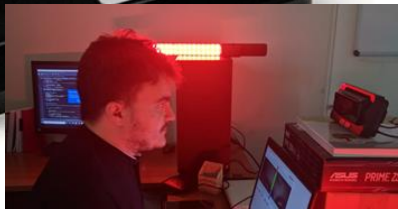
# Iris sensing

- **Iris can be acquired reliably only at a close distance**

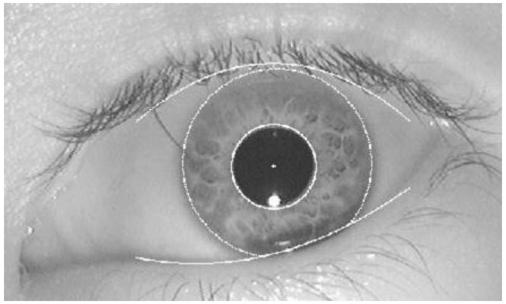



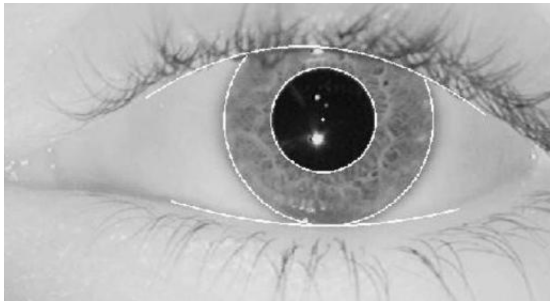



# Iris pre-processing

- **Iris diameter is typically between 200 and 250 pixels from a distance of about 15 cm, using a standard lens and video camera with video rate capture**

- **Goal: isolation of iris from the rest of the image**





- **Problem:**
  – The eyelashes and eyelids may partially overlap the pupillary zone
  – The larger zone must be chosen between the acquired images

# Iris segmentation methods

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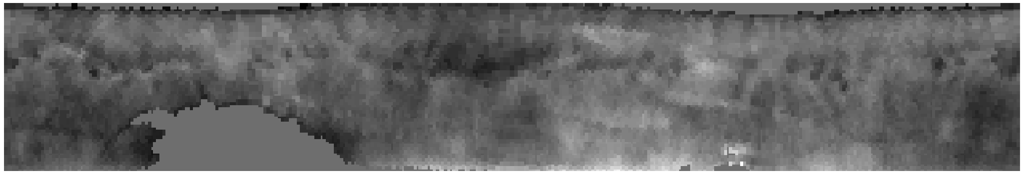
- **Circular Hough Transform (CHT)**
  - Used for detecting circles in an image
- **Integrodifferential operator (Daugman, 2009)**
  - the illumination difference between inside and outside of pixels in iris edge circle is maximum
  - the difference in values of pixels gray level in iris circle is higher than any other circles in images
  - this fact turns to color of iris and color of sclera, which is the white region outside of iris
- **Active Contour techniques (similar to ASM in faces)**
- **Convolutional Neural Networks**

# Iris feature extraction



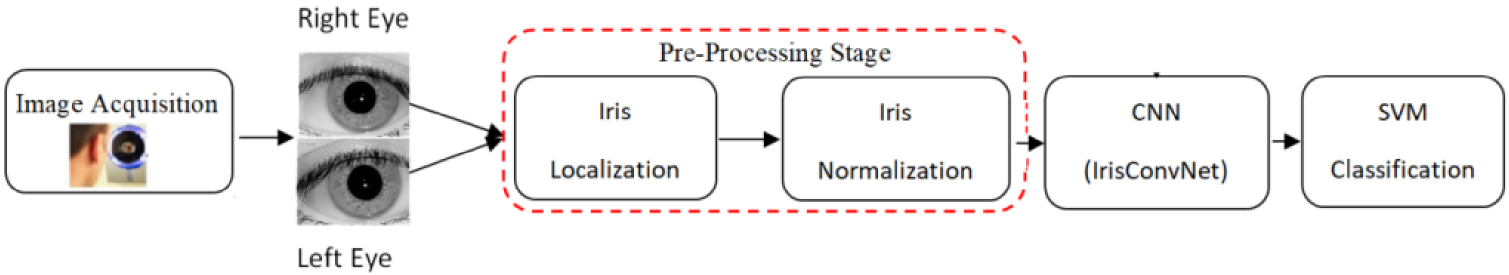
- **Filterbank based filtering followed by statistics computation**
  - Gabor filters: Daugman
- **Textural description by set of histograms**
  - LBP, BSIF, LPQ
- **Deep feature extraction by CNNs ➔ embeddings**
- **The so-called «iris-code» is the *feature vector* derived from the chosen approach**

# Iris match

- **The original iris-code transformed the unfolded iris in a bit string**
  - Hamming distance

- **In general, the usual distances can be adopted according to the hypothesis on the feature space**
  - L1: Hellinger
  - L2: Euclidean or Cosine

- **Deep networks may also embed the «distance» definition**

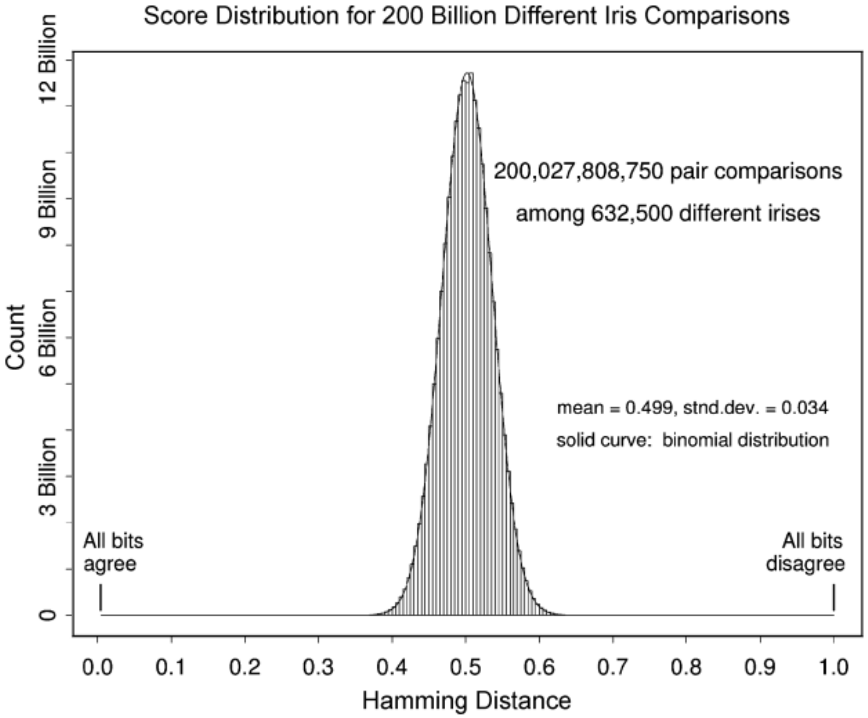


# How about randomness?

- **Daugman studied the randomness of «his» iris-code by a robust statistical verification hypothesis test**

- **It is informative to calculate the significance of any observed HD (Hamming Distance) matching score, in terms of the likelihood that it could have arisen by chance from two different irises**

- **These probabilities give a confidence level associated with any recognition decision**



# How about uniqueness?

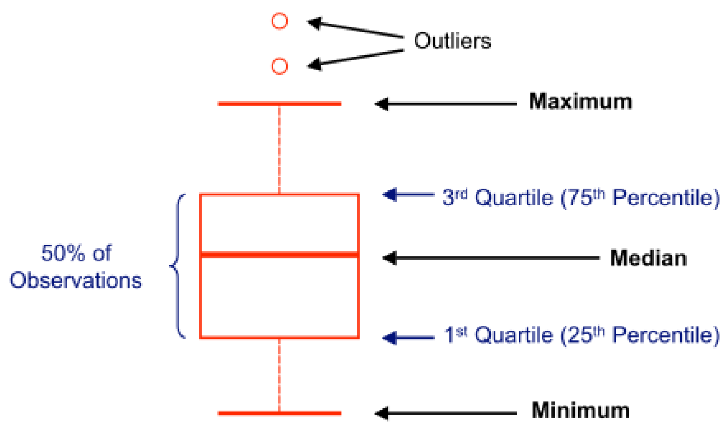
- **Daugman also studied the uniqueness of «his» iris-code**
- **In this case, the estimated $P = FMR$ was very low according to the observed user population**
- **According to the Daugman's findings, the probability of making at least one false match when searching a database containing N different patterns is:**

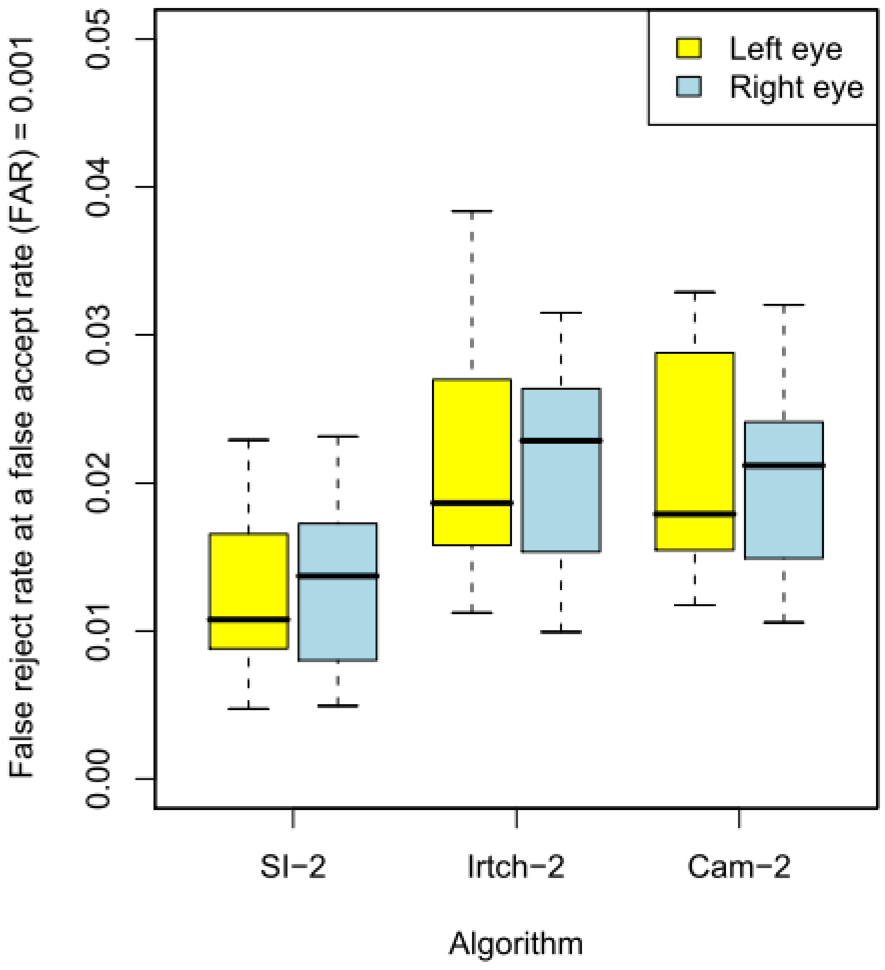
$$P_N = 1 - (1 - P)^N$$

| *HD Criterion* | *Observed False Match Rate* |
|---|---|
| 0.220 | 0 (*theor: 1 in* $5 \times 10^{15}$) |
| 0.225 | 0 (*theor: 1 in* $1 \times 10^{15}$) |
| 0.230 | 0 (*theor: 1 in* $3 \times 10^{14}$) |
| 0.235 | 0 (*theor: 1 in* $9 \times 10^{13}$) |
| 0.240 | 0 (*theor: 1 in* $3 \times 10^{13}$) |
| 0.245 | 0 (*theor: 1 in* $8 \times 10^{12}$) |
| 0.250 | 0 (*theor: 1 in* $2 \times 10^{12}$) |
| 0.255 | 0 (*theor: 1 in* $7 \times 10^{11}$) |
| 0.262 | 1 in 200 billion |
| 0.267 | 1 in 50 billion |
| 0.272 | 1 in 13 billion |
| 0.277 | 1 in 2.7 billion |
| 0.282 | 1 in 284 million |
| 0.287 | 1 in 96 million |
| 0.292 | 1 in 40 million |
| 0.297 | 1 in 18 million |
| 0.302 | 1 in 8 million |
| 0.307 | 1 in 4 million |
| 0.312 | 1 in 2 million |
| 0.317 | 1 in 1 million |

# Iris Challenge Results (NIST)

- Carried out in 2006
- Cambridge (Cam-2), Iritech (IrTch-2) and Sagem-Iridian (SI-2) systems
- 29,056 images of 240 subjects
- How to read the plots:



# Experiments carried out at → sAIfer Lab

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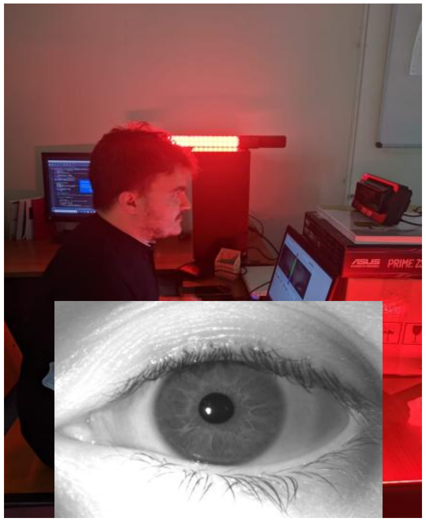

- **30 people, 10 images per eye (left and right)**

- **Four different lighting conditions**
  - diffused artificial light
  - artificial light ( illumination from the right )
  - artificial light ( illumination from the left )
  - diffused natural light

| Method | | FRR | ACC |
|---|---|---|---|
| BSIF | Single eye (left) | 7.48% | 99.63% |
| | Score level fusion | 0.63% | 99.83% |
| Gabor | Single eye (left) | 4.90% | 99.75% |
| | Score level fusion | 0.36% | 99.91% |

# Homework

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- **Download some iris segmentation algorithms from internet or implement some basics one as CHT**
- **Download some iris images and see differences in segmentation results when using your own cellphone to capture**
- **Implement the IrisCode process using the same FingerCode one (e.g. using Gabor's filters)**
- **Assess the genuine and impostor distributions using several distances according to different feature space hypothesis**
- **Are your results compliant with what was reported by the state-of-the-art?**

# Iris Presentation Attack Detection

# Iris presentation attacks

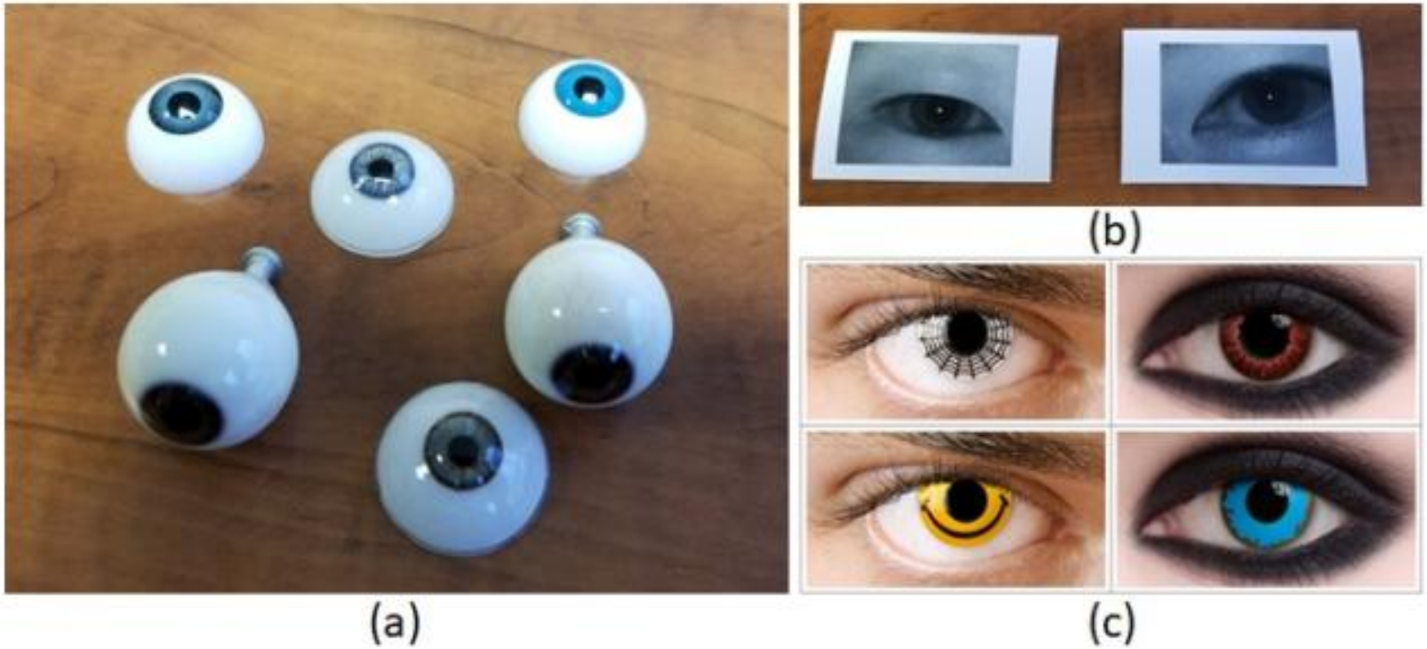
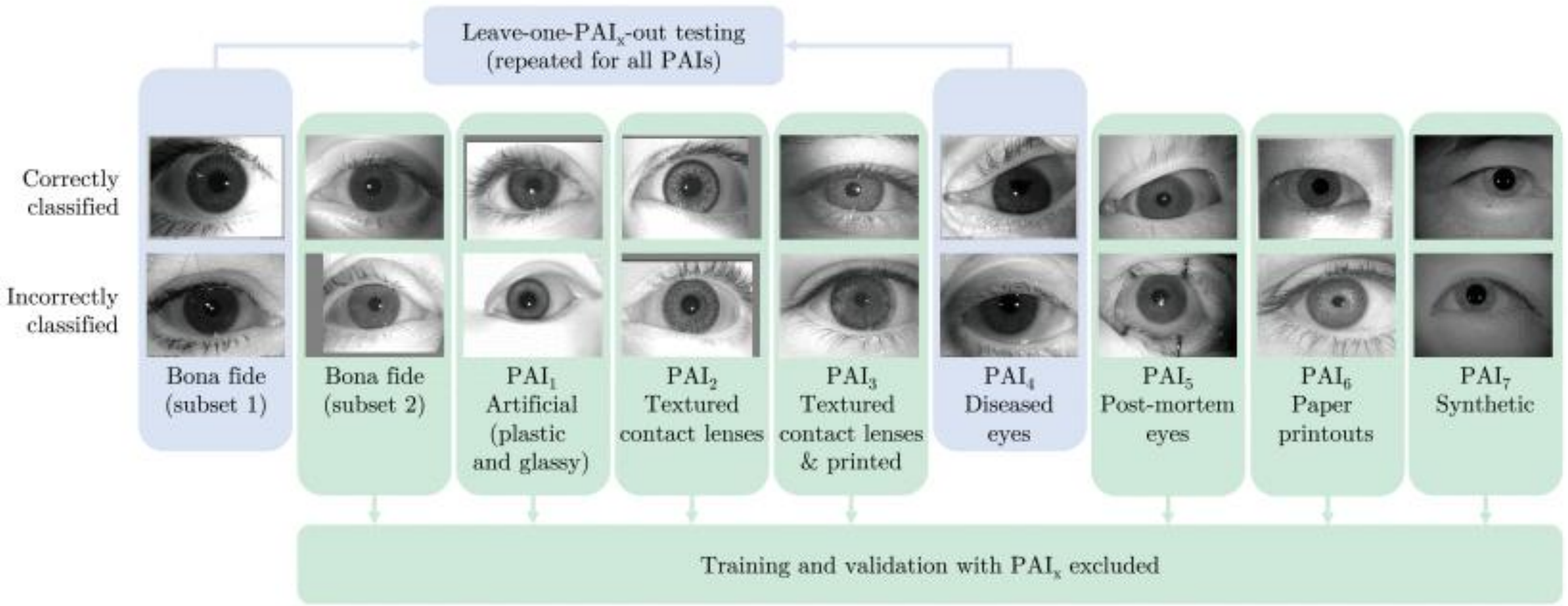



Figure 1. Examples of PAI used to launch presentation attacks (PAs) on the iris modality: (a) plastic eyes, (b) printed images, and (c) cosmetic contacts [20].

https://arxiv.org/abs/2311.12773

# A more comprehensive set of PAIs






https://ieeexplore.ieee.org/document/10122228

# Data sets

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| Image Type | Combined Dataset: Contributing Dataset | Combined Dataset: # of Samples | Total # of Samples | LivDet-Iris 2020 # of Samples |
|---|---|---|---|---|
| Bona fide | ATVS-FIr [22] | 800 | 399,053 | 5,331 |
| | BERC_IRIS_FAKE [23] | 2,776 | | |
| | CASIA-Iris-Thousand [24] | 19,952 | | |
| | CASIA-Iris-Twins [24] | 3,181 | | |
| | Disease-Iris v2.1 [25] | 255 | | |
| | ETPAD v2 [26] | 400 | | |
| | IIITD Contact Lens Iris [27] | 13 | | |
| | IIITD Combined Spoofing Database [28] | 4,531 | | |
| | LivDet-Iris Clarkson 2015 [11] | 813 | | |
| | LivDet-Iris Warsaw 2015*** [11] | 36 | | |
| | LivDet-Iris Clarkson 2017 [14] | 3,949 | | |
| | LivDet-Iris IIITD-WVU 2017 [14] | 2,944 | | |
| | LivDet-Iris Warsaw 2017*** [14] | 5,167 | | |
| | Notre Dame (previously released data)* | 217,712 | | |
| | Notre Dame (new data released with this paper)** | 136,524 | | |
| Artificial | BERC_IRIS_FAKE [23] | 80 | 277 | 541 |
| | Notre Dame (previously released data)* | 91 | | |
| | Notre Dame (new data released with this paper)** | 106 | | |
| Textured contact lenses | BERC_IRIS_FAKE [23] | 140 | 27,372 | 4,336 |
| | IIITD Contact Lens Iris [27] | 3,420 | | |
| | LivDet-Iris Clarkson 2015 [11] | 1,107 | | |
| | LivDet-Iris Clarkson 2017 [14] | 1,881 | | |
| | LivDet-Iris IIITD-WVU 2017 [14] | 1,700 | | |
| | Notre Dame (previously released data)* | 2,751 | | |
| | Notre Dame (new data released with this paper)** | 16,373 | | |
| Textured contact lenses & printed | LivDet-Iris IIITD-WVU 2017 [14] | 1,899 | 1,899 | — |
| Diseased | Disease-Iris v2.1 [25] | 1,537 | 1,537 | — |
| Post-mortem | Post-Mortem-Iris v3.0 [29] | 2,259 | 2,259 | 1,094 |
| Paper printouts | ATVS-FIr [22] | 800 | 16,393 | 1,049 |
| | BERC_IRIS_FAKE [23] | 1,600 | | |
| | IIITD Combined Spoofing Database [28] | 1,371 | | |
| | LivDet-Iris Clarkson 2015 [11] | 1,745 | | |
| | LivDet-Iris Warsaw 2015*** [11] | 20 | | |
| | LivDet-Iris Clarkson 2017 [14] | 2,250 | | |
| | LivDet-Iris IIITD-WVU 2017 [14] | 1,766 | | |
| | LivDet-Iris Warsaw 2017*** [14] | 6,841 | | |
| Synthetic | CASIA-Iris-Syn V4 [30] | 10,000 | 10,000 | — |
| Displayed on e-ink device | — | — | — | 81 |
| **All combined** | | | **458,790** | **12,432** |

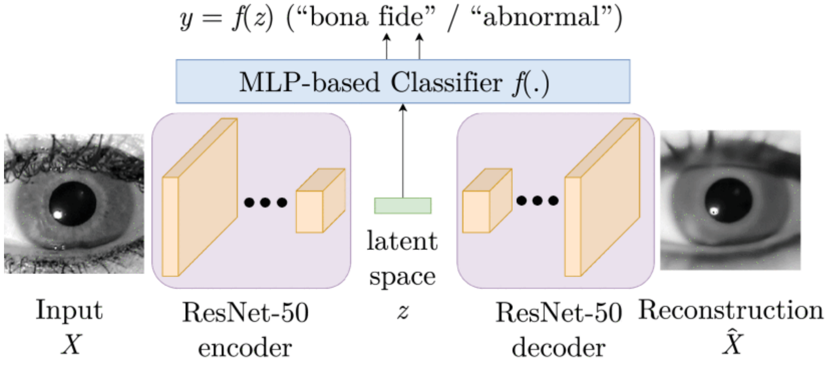
* University of Notre Dame has released so far 238,266 iris images (bona fide and various PAIs). All previously released data sets are listed, and a copy can be requested at https://cvrl.nd.edu/projects/data

**We release 153,003 new iris images with this paper, previously never published. This contributes to 391,269 of all iris samples (bona fide and PAIs) released by Notre Dame to date. A copy of this new dataset will be possible to be requested at cvrl@nd.edu upon acceptance of this paper.

*** According to http://zbum.ia.pw.edu.pl/EN/node/46 (accessed on Feb. 8, 2023), this dataset is no longer available due to EU's GDPR regulation.

# Methods

- **Domain-Aware Convolutional Neural Network**
- **Regional Features-Based Iris PAD**
  - BSIF etc.
- **D-NetPAD**
  - DenseNet backbone
  - Attention-block can be added
- **Variational Autoencoder**



# Current performances

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- **450,000 images representing authentic iris and seven types of presentation attack instrument (PAI)**
- **Available Open-Source Algorithms**

| Method | CCR ± 1$\sigma$ | APCER ± 1$\sigma$ | BPCER ± 1$\sigma$ |
|---|---|---|---|
| **RegionalPAD** | 94.24% ± 0.06% | 33.41% ± 0.52% | 1.65% ± 0.05% |
| **DACNN** | 98.81% ± 0.03% | 5.36% ± 0.67% | 0.57% ± 0.08% |
| **D-NetPAD1** | 98.53% ± 1.25% | 11.26% ± 9.63% | **0.00% ± 0.00%** |
| **D-NetPAD2** | **99.92% ± 0.01%** | **0.18% ± 0.05%** | 0.06% ± 0.01% |
| **OSIPAD** | 99.80% ± 0.26% | 0.85% ± 1.16% | 0.10% ± 0.12% |
| **VAEPAD** | 99.26% ± 0.03% | 3.15% ± 0.39% | 0.38% ± 0.05% |
| **Ensemble (poly)** | 99.97% ± 0.04% | 0.14% ± 0.19% | **0.01 % ± 0.02 %** |
| **Ensemble (RBF)** | **99.97% ± 0.04%** | **0.09% ± 0.13%** | 0.02% ± 0.02% |
| **Ensemble (sigmoid)** | 98.47% ± 0.28% | 5.27% ± 0.83% | 0.97% ± 0.20% |

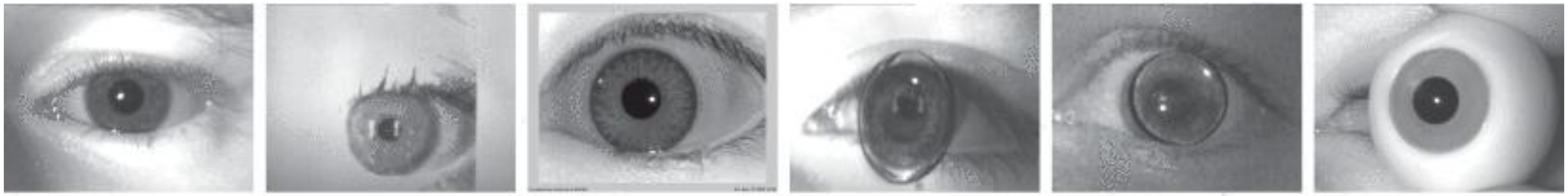
A. Boyd, J. Speth, L. Parzianello, K. W. Bowyer and A. Czajka, Comprehensive Study in Open-Set Iris Presentation Attack Detection, *IEEE Transactions on Information Forensics and Security*, vol. 18, pp. 3238-3250, 2023, doi: 10.1109/TIFS.2023.3274477

# Iris LivDet 2023

sAIfer Lab
Joint lab on Safety and Security of AI

- **Part 1: "Algorithms-Self-Tested"**
  - self-evaluation (that is, done by competitors on the sequestered and never-published-before test dataset) incorporating a large number of ISO-compliant live and fake samples. Samples included irises synthesized by modern Generative Adversarial Networks-based models (StyleGAN2 and StyleGAN3) and near-infrared pictures of various artefacts simulating physical iris presentation attacks.
- **Part 2: "Algorithms-Independently-Tested“**
  - evaluation of the software solutions submitted by the competitors and performed by the organizers on a sequestered dataset similar in terms of attack types to the one used in Part 1.
- **Part 3: "Systems“**
  - systematic testing of submitted iris recognition systems based on physical artifacts presented to the sensors.
- **Results in: https://ieeexplore.ieee.org/document/10448637**

# LivDet data



(a) Bona fide Iris | (b) Paper printout | (c) Cosmetic contact lens on bona fide eye | (d) Cosmetic contact lens on printed eye | (e) Eye dome on the printed eye | (f) Doll eye

(g) Electronic display | (h) Cosmetic contact lens on the doll eye | (i) Synthetic Iris (Low Quality) | (j) Synthetic Iris (Medium Quality) | (k) Synthetic Iris (High Quality)

| Class | Presentation Attack Instruments (PAI) | Sample Count | Sensor |
|---|---|---|---|
| Bona fide | N/A | 6500 | LG 4000, AD 100, Iris ID iCAM7000 |
| PAI | Printed Eyes (PE) | 522 | Iris ID iCAM7000 |
| PAI | Textured Contact Lens (CL) | 4 | LG 4000, AD 100, Iris ID iCAM7000 |
| PAI | Electronic Display (ED) | 40 | Iris ID iCAM7000 |
| PAI | Fake/Prosthetic/Printed Eyes with Add-Ons (FP) | 266 | Iris ID iCAM7000 |
| PAI | Synthetic Iris - Low Quality (LQ) | 2000 | N/A |
| PAI | Synthetic Iris - Medium Quality (MQ) | 2000 | N/A |
| PAI | Synthetic Iris - High Quality (HQ) | 2000 | N/A |

# LivDet results



| Algorithm | APCER (%) Level A PE | Level B F/P | CL | ED | Level C - Synth LQ | MQ | HQ | Overall Metrics (%) BPCER | APCER1 | APCER2 | ACER (%) ACER1 | ACER2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **LivDet-Iris 2023 Competing Algorithms** | | | | | | | | | | | | |
| Fraunhofer IGD | 0.57 | 0.00 | 0.00 | 0.00 | 30.00 | 51.35 | 52.50 | 35.40 | 39.22 | 19.20 | <u>**37.31**</u> | 27.30 |
| BUCEA Algo1 | 22.80 | 1.50 | 0.00 | 0.00 | 95.55 | 94.40 | 94.85 | 0.14 | 85.16 | 44.16 | 42.65 | <u>**22.15**</u> |
| BUCEA Algo3 | 29.12 | 7.14 | 0.00 | 2.50 | 99.20 | 99.10 | 99.35 | 0.18 | 89.64 | 48.06 | 44.91 | 24.12 |
| BUCEA Algo4 | 38.70 | 7.52 | 0.00 | 15.00 | 98.65 | 99.25 | 99.45 | 0.60 | 90.37 | 51.22 | 45.49 | 25.91 |
| BUCEA Algo2 | 51.34 | 25.94 | 0.00 | 27.50 | 97.20 | 98.15 | 97.95 | 0.78 | 90.94 | 56.87 | 45.86 | 28.82 |
| HDA-IDVC Algo1 | 1.53 | 1.13 | 0.00 | 0.00 | 67.15 | 66.00 | 67.80 | 42.34 | 58.98 | 29.09 | 50.66 | 35.71 |
| HDA-IDVC Algo2 | 2.30 | 0.00 | 0.00 | 0.00 | 62.55 | 63.95 | 65.00 | 49.92 | 56.23 | 27.69 | 53.07 | 38.80 |
| HDA-IDVC Algo3 | 2.30 | 0.00 | 0.00 | 0.00 | 62.55 | 63.95 | 65.00 | 49.92 | 56.23 | 27.69 | 53.07 | 38.80 |
| **LivDet-Iris 2023 Baseline Algorithms** | | | | | | | | | | | | |
| DenseNet (vanilla) | 0.00 | 0.00 | 0.00 | 0.00 | 88.45 | 95.85 | 97.25 | 0.34 | 82.41 | 40.22 | 41.37 | 20.28 |
| DenseNet (augs) | 0.00 | 0.38 | 0.00 | 0.00 | 96.50 | 97.60 | 98.85 | 0.17 | 85.76 | 41.90 | 42.97 | 21.04 |
| DenseNet (synths) | 0.00 | 0.00 | 0.00 | 0.00 | 0.10 | 0.50 | 0.15 | 0.46 | 0.22 | 0.11 | 0.34 | 0.28 |
| DenseNet (both) | 0.38 | 0.38 | 0.00 | 0.00 | 0.35 | 0.55 | 0.15 | 0.26 | 0.35 | 0.26 | 0.31 | 0.26 |
| Inception (vanilla) | 0.19 | 1.13 | 0.00 | 0.00 | 96.10 | 98.80 | 99.55 | 0.11 | 86.24 | 42.25 | 43.18 | 21.18 |
| Inception (augs) | 0.00 | 0.38 | 0.00 | 0.00 | 87.00 | 93.65 | 94.35 | 0.31 | 80.51 | 39.34 | 40.41 | 19.83 |
| Inception (synths) | 0.00 | 0.00 | 0.00 | 0.00 | 0.10 | 0.15 | 0.05 | 0.45 | 0.09 | 0.04 | 0.27 | 0.25 |
| Inception (both) | 0.00 | 0.00 | 0.00 | 0.00 | 0.80 | 0.15 | 0.05 | 0.31 | 0.29 | 0.14 | 0.30 | 0.23 |
| ResNet (vanilla) | 0.00 | 0.00 | 0.00 | 0.00 | 95.85 | 97.45 | 98.00 | 0.15 | 85.26 | 41.61 | 42.71 | 20.88 |
| ResNet (augs) | 1.34 | 0.00 | 0.00 | 2.50 | 97.75 | 98.60 | 99.45 | 0.32 | 86.70 | 42.81 | 43.51 | 21.56 |
| ResNet (synths) | 0.00 | 0.00 | 0.00 | 0.00 | 0.20 | 0.20 | 0.10 | 0.43 | 0.15 | 0.07 | 0.29 | 0.25 |
| ResNet (both) | 0.00 | 0.38 | 0.00 | 0.00 | 0.40 | 1.80 | 1.25 | 0.12 | 1.02 | 0.55 | 0.57 | 0.33 |
| ViT (vanilla) | 0.00 | 0.00 | 0.00 | 0.00 | 97.35 | 99.00 | 99.55 | 0.15 | 86.61 | 42.27 | 43.38 | 21.21 |
| ViT (augs) | 1.53 | 0.38 | 0.00 | 0.00 | 97.40 | 99.35 | 99.65 | 0.05 | 86.89 | 42.62 | 43.47 | 21.33 |
| ViT (synths) | 0.19 | 0.00 | 0.00 | 0.00 | 2.70 | 1.90 | 0.95 | 0.46 | 1.64 | 0.82 | 1.05 | 0.64 |
| ViT (both) | 0.77 | 0.00 | 0.00 | 0.00 | 11.00 | 7.95 | 4.90 | 0.18 | 7.04 | 3.52 | 3.61 | 1.85 |
| StyleGAN2 (LQ) | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 100.00 | 0.00 | 0.00 | 50.00 | 50.00 |
| StyleGAN2 (MQ) | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 100.00 | 0.00 | 0.00 | 50.00 | 50.00 |
| StyleGAN2 (HQ) | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 100.00 | 0.00 | 0.00 | 50.00 | 50.00 |
| StyleGAN3 (LQ) | 35.25 | 23.68 | 25.00 | 32.50 | 6.00 | 8.05 | 7.75 | 96.17 | 10.20 | 19.75 | 53.19 | 57.96 |
| StyleGAN3 (MQ) | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 100.00 | 0.00 | 0.00 | 50.00 | 50.00 |
| StyleGAN3 (HQ) | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 100.00 | 0.00 | 0.00 | 50.00 | 50.00 |
| Xception (Xent) | 11.69 | 62.41 | 0.00 | 7.50 | 87.15 | 95.90 | 96.10 | 2.63 | 85.07 | 51.54 | 43.85 | 27.08 |
| Xception (CYBORG) | 18.20 | 40.98 | 25.00 | 17.50 | 77.55 | 87.40 | 88.30 | 7.79 | 77.23 | 50.70 | 42.51 | 29.25 |
| **LivDet-Iris 2023 Human Examination** | | | | | | | | | | | | |
| Human subjects | 30.12 | 7.06 | 20.41 | 17.57 | 33.19 | 50.25 | 51.14 | 38.38 | 40.98 | 29.96 | 39.68 | 34.17 |

Unbalanced accuracy

Too many errors!

# Homework

sAIfer Lab
Joint lab on Safety and Security of AI

- **Explore if iris-code has also liveness detection potential**
- **Download some iris presentation attack detection algorithms**
- **Download some presentation attack detection data sets with several captured PAIs**
- **Compare official results with yours**

# Synthetic iris generation

- **Proposed since from 2017 by Drozdowski *et al.***

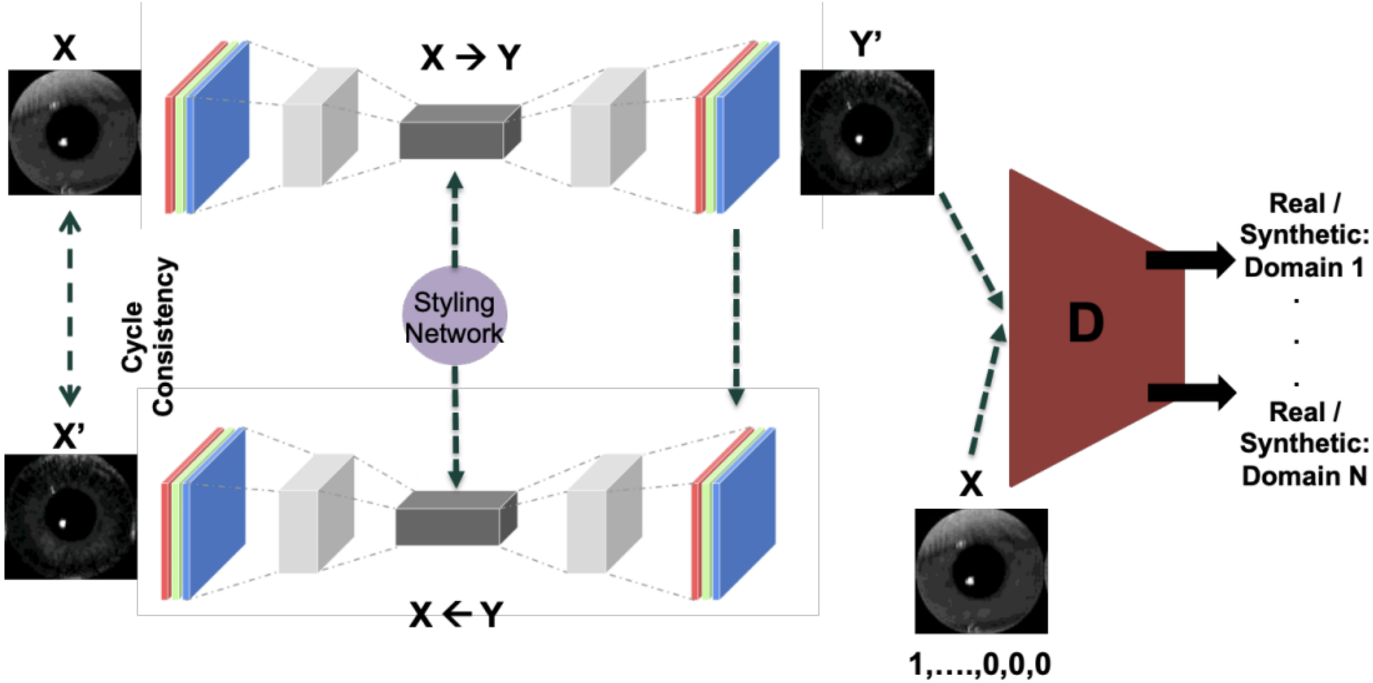
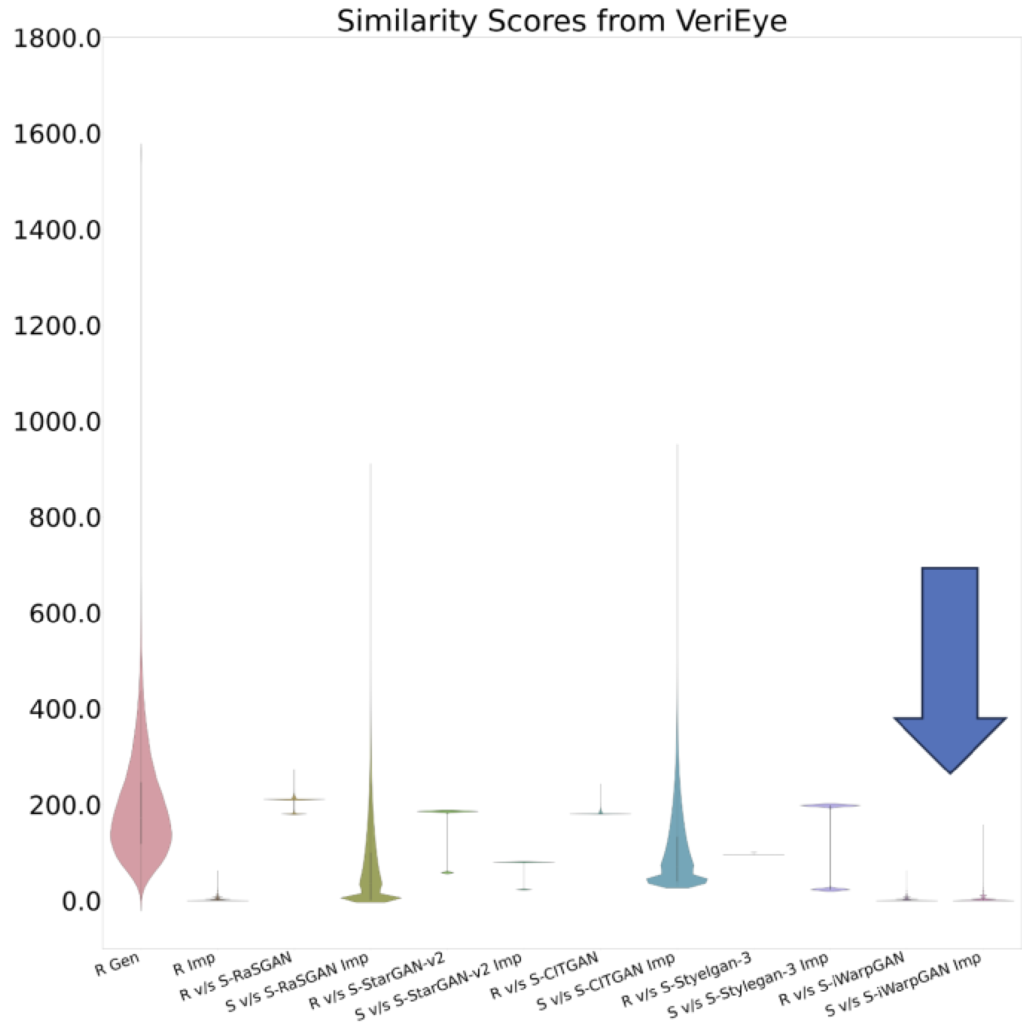


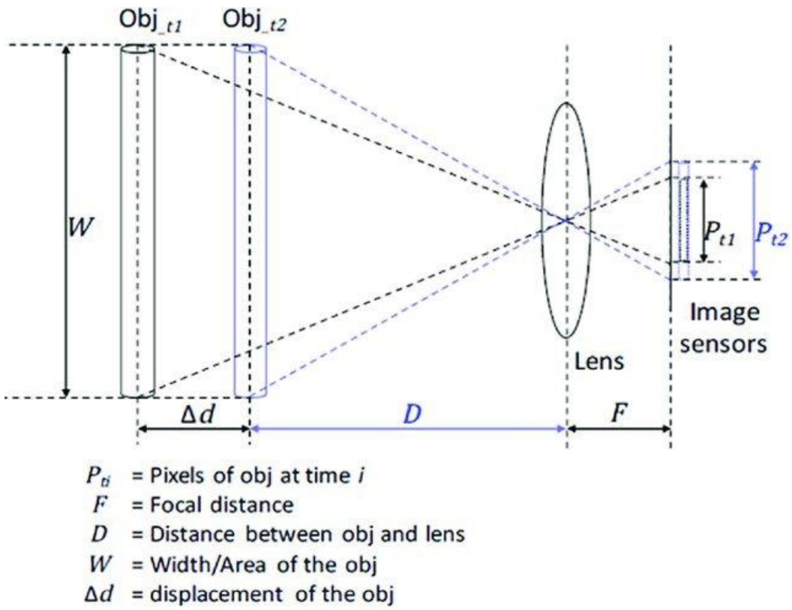

Figure 7. Yadav and Ross [90] proposed CIT-GAN where the generator aims to translate given input image into an image from reference domain. This translation is facilitated by a Style Network that takes reference image as input and actively learns to generate style code that clearly indicates the style properties of reference domain

# Uniqueness test



# Our challenge

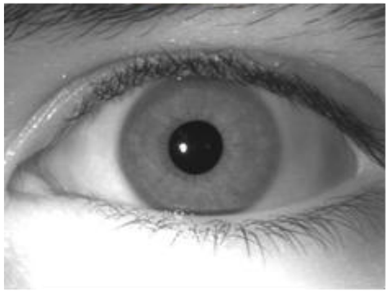
- **Iris recognition at a distance**
  - More than 40 cm
- **Problems**
  - Loss of details
  - Non-linear distortion
- **If the focal length of the lens is reasonably long (POV < 120°):**
  - The camera follows the laws of perspective transformation and linear projection
  - Radial distortions generated by the width of the field angle can be neglected



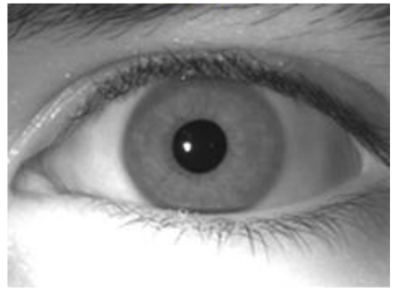

Fan, Jinlong, et al. "Wide-angle image rectification: A survey." *International Journal of Computer Vision* 130.3 (2022): 747-776.

# Simulation of the details' loss effect

33 cm



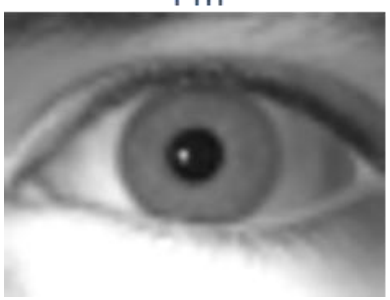
1 m



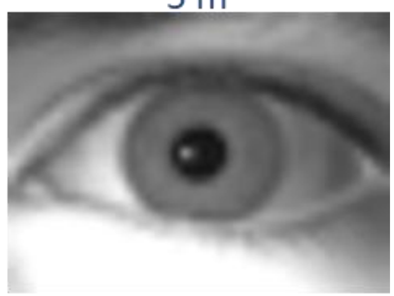
2 m

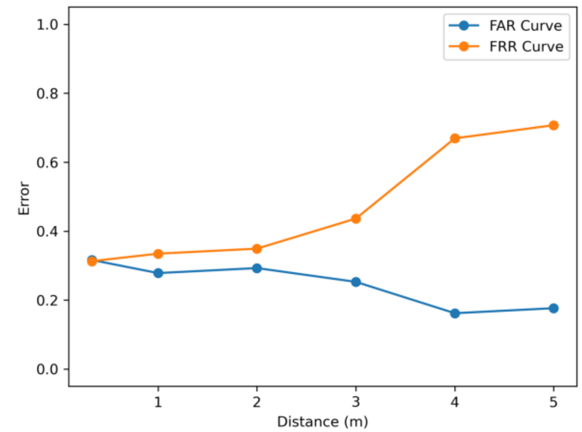
3 m

4 m

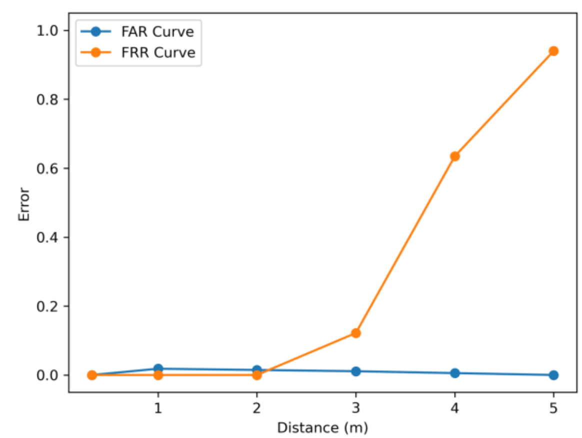


5 m





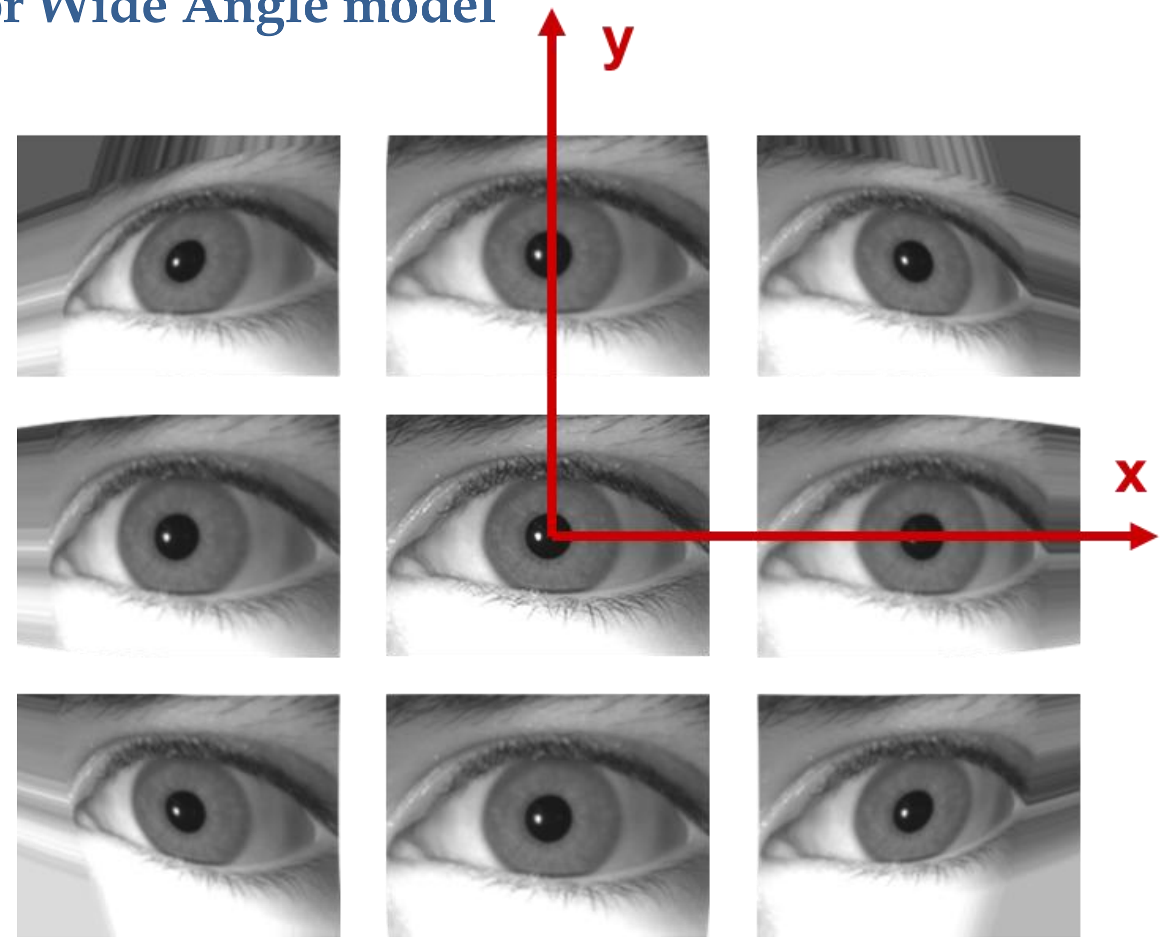

Daugman




VeriEye SDK

# Next step: non-linear distortions

- **Fisheye or Wide Angle model**




THIS STORY HAS YET TO BE TOLD

# That's all for today… what's next?

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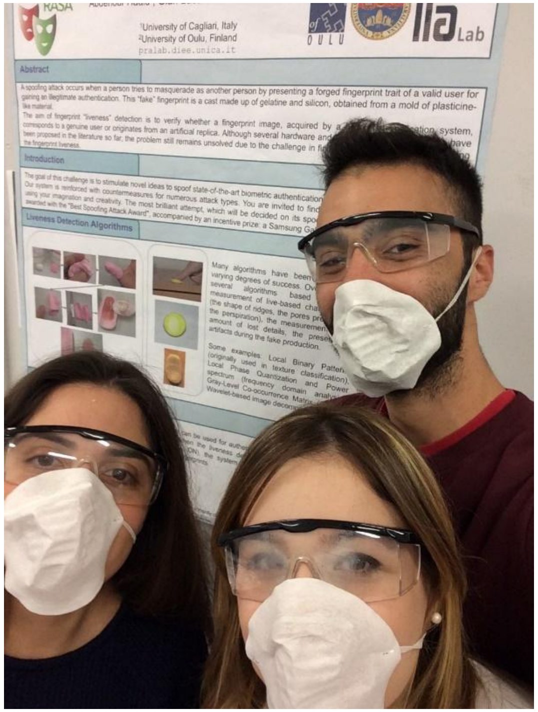
- **Multi-modal biometric systems**
  - properties
  - pros and cons
- **Fusion approaches**
  - Feature-level fusion
  - Measurement-level fusion
  - Decision-level modern challenge
- **Fusion architectures**
  - Parallel
  - Serial

# Thank you for listening!





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